Soils of India, Types, Significance

Soils in India include a diverse range of types such as alluvial, black, red, laterite, arid, and forest soils. Check out more Soils of India, Types and Significance.

By Vajiram Editor - Apr 9, 2026, 10:47 IST

Table of Contents

The soils of India are diverse and shaped by climate, parent material, topography, biological activity, and time. The major soil types include alluvial, black, red, laterite, desert, forest, peaty, and saline soils, each with distinct characteristics and agricultural significance. Alluvial soil is the most fertile and widespread, crucial for agriculture, while black soil is ideal for cotton.

However, soil degradation affects 30% of India's land, caused by unsustainable farming, poor water management, deforestation, and urbanization. Conservation efforts are essential to maintain soil health, ensuring food security and economic stability.

Enquire Now

Soils of India

India's diverse geography and climate have resulted in the formation of a wide variety of soils in India, each playing a crucial role in the agricultural landscape. **Soil formation** is a complex process influenced by several factors, including climate, parent material, topography, biological activity, and time.

- The type of **parent material**, such as igneous or sedimentary rocks, determines the soil's composition.
- **Climate**, particularly temperature and rainfall, affect weathering rates and soil fertility, with high rainfall regions experiencing leaching and arid zones leading to less fertile soils.
- **Topography** influences soil moisture and vegetation, while biological activity, including the decomposition of organic matter, contributes to soil fertility.

Over time, these factors interact, gradually forming distinct soil profiles.

Soils of India Classification

The soils of India are classified based on various criteria, including their physical and chemical properties, as well as their agricultural significance. The **Indian Council of Agricultural Research (ICAR)** has developed a comprehensive classification system that categorizes Indian soils into eight major groups: alluvial, black, red, laterite, desert (arid), forest, peaty, and saline soils.

Alluvial Soil

Alluvial soil is the most extensive and agriculturally significant soil type in India. Covering about 15 lakh square kilometres or **45.6%** of the country's total land area, it forms the foundation of India's agricultural wealth and supports a majority of the population.

- **Formation and Composition:** Alluvial soils are formed by the **deposition of sediments** (alluvium) by rivers, winds, glaciers, and sea waves. They are rich in humus and nutrients like potash but often lack phosphorus.
  - Their fertility is further enhanced as they are renewed annually by river floods.
- **Types of Alluvial Soil:** Alluvial soils are categorized into two types:
- Old alluvium (**Bhangar**) is clayey, sticky, and darker with lime nodules.
- New alluvium (**Khadar**) is lighter and deposited in floodplains and deltas.
- **Distribution:** **Great Northern Plain** (from Punjab to West Bengal and Assam), the lower valleys of the **Narmada** and **Tapti** rivers, the east coast deltas, and the northern **Gujarat plains** through a narrow corridor in Rajasthan.
- **Characteristics:** Alluvial soils range in texture from sandy loam to clay and vary in colour from light grey to ash grey based on maturity.
  - They are more loamy and clayey in the lower Ganga plain and Brahmaputra valleys with sand content decreasing from west to east.

Enquire Now

- **Crops:** Rice, Wheat, Sugarcane, Cotton, Tobacco, Pulses, Oilseeds and Jute.

Black Soil (Regur Soil)

The black soil, also known as **Regur** or **Black Cotton Soil**, is primarily found in the **Deccan Plateau**, covering approximately 5.46 lakh square kilometres (**16.6%** of India's total geographical area).

- **Formation and Composition:** Black soil, formed from **fissure volcanic rock,** is ideal for cotton cultivation, earning it the name "black cotton soil."
  - It contains alumina, iron oxide, lime, magnesium carbonates, and poor nitrogen, phosphorus, and organic matter content.

- **Characteristics**: Black soils are highly **argillaceous**, with over 62% clay, and are excellent at retaining moisture.
  - They have a black colour due to **titaniferous magnetite** or **iron** and are ideal for cotton cultivation despite being deficient in essential nutrients.

- **Crops:** Cotton, sugarcane, tobacco, millets (Jowar, sorghum), wheat, oilseeds (groundnut, linseed), pulses (lentils, chickpeas), citrus fruits, sunflowers, turmeric.

Red soils occupy about 3.5 lakh sq km, approximately **10.6%** of India's total land area. They are widely distributed across the country, notably in parts of south Bihar, West Bengal, Uttar Pradesh, Aravallis, eastern Rajasthan, and parts of Assam, Nagaland, Manipur, Mizoram, Tripura, and Meghalaya.

- **Formation and Composition:** Red soil forms from the weathering of ancient crystalline rocks, with its red colour due to iron oxide. It is rich in iron but poor in nitrogen, phosphorus, and organic matter content.
- **Characteristics:** Red soils have textures from sandy to loamy, with moderate fertility, suited for dry farming but require irrigation and fertilizers for enhanced productivity.
- **Crops:** Cotton, groundnut, tobacco, and pulses, though less suitable for water-intensive crops due to limited moisture retention.

Laterite Soil

Laterite soils, named after the Latin word "later" meaning **brick** which hardened upon exposure. Found in regions like South Maharashtra, the Western Ghats in Kerala and Karnataka, parts of Assam, Tamil Nadu, and Western West Bengal (especially in Birbhum district), as well as parts of the Eastern Ghats.

- **Formation and Composition:** Laterite soils form under high heat and rainfall, causing intense mineral leaching.
  - They are rich in iron, aluminium, and manganese, giving them reddish to brown hues.

- **Characteristics**: Laterite soils are clayey, compact, and poorly drained, m[illegible] for farming without proper management.

Enquire Now

  - They are poor in essential nutrients like lime, nitrogen, and phosphorus, requiring fertilisers and soil amendments.

- **Crops:** Cashews, tea, coffee, tapioca, rubber, and tropical fruits.
- **Other Uses**: Laterite soils are often used in construction, particularly for making bricks, and roads, and as a building material due to their hardened texture after exposure to air.

Desert (Arid) Soil

Desert soils are primarily found in the arid regions of Rajasthan, Haryana, South Punjab, and parts of Gujarat, as well as near the Indus River and the Aravallis. The **Thar Desert** is one of the largest stretches of desert soil in India, covering vast areas of northwestern India.

- **Composition**: Desert soils contain high amounts of soluble salts and calcium carbonate, with calcium increasing in deeper layers, making them less fertile.
- **Distribution**: Desert soils cover about 1.42 lakh square kilometres, which is approximately **4.32%** of the total land area of India.
  - These soils are concentrated in the northwestern and western parts of the country.
- **Characteristics:** Saline soils, with low rainfall and poor leaching, are challenging for cultivation without irrigation.
  - Their sandy texture, low organic matter, and poor moisture retention hinder fertility and plant growth.
- **Crops:** Only drought-resistant and salt-tolerant crops likebarley, cotton, wheat, millets, and pulses are grown in desert soils.

Forest Soil

Forest soils are predominantly found in the Himalayan region, Western Ghats, and parts of northeastern states like Assam, Nagaland, and Mizoram. These soils are also seen in tropical and subtropical areas, influenced by the type of vegetation and climate.

- **Composition:** Forest soils are rich in organic matter, including humus from decomposed plant material, and are high in essential nutrients like nitrogen, phosphorus, and potassium.
- **Characteristics:** Forest soils vary in texture from sandy loam to clayey, with colours ranging from light to dark brown due to high organic content.
  - They are fertile but **acidic**, and deforestation can degrade soil quality, leading to erosion and reduced productivity.
- **Crops:** Tea, coffee, spices, wheat, maize, barley etc.

Enquire Now

Peaty Soil

Peaty and marshy soils form in humid regions with excess moisture and waterlogging, slowing plant decomposition. These soils accumulate organic matter under stagnant water and are found in wetland areas like Kottayam and Alappuzha in Kerala, coastal Orissa, Tamil Nadu, Sunderbans in West Bengal, and parts of Bihar and Uttaranchal.

- **Formation and Composition:** Peaty soils are black due to accumulated decomposed plant material, rich in organic matter and nitrogen, but often lacking in phosphorus and potassium, requiring additional fertilization for crops needing more minerals.
- **Characteristics:** Peaty soils are heavy, spongy, and moisture-retentive, while marshy soils are rich in organic matter and suited for paddy cultivation.
  - Both are naturally fertile but highly acidic, requiring proper water management for optimal growth.
- **Crops:** Cabbage, broccoli, peas, beans, carrots, radishes, potatoes, lettuce, spinach.

Saline Soil

Saline soils are found in areas like Andhra Pradesh, Karnataka, Bihar, Uttar Pradesh, Haryana, Punjab, Rajasthan, Maharashtra, and parts of Gujarat covering about 68,000 sq km. The accumulation of salts renders soil infertile, with 1.25 million hectares in Uttar Pradesh, 1.21 million hectares in Punjab, and areas around the Gulf of Khambhat in Gujarat affected.

- **Formation**: Saline and alkaline soils are formed in dry regions due to the accumulation of salts like sodium, magnesium, and calcium, primarily through evaporation and weathering of rock and mineral fragments.
- **Characteristics:** Saline and alkaline soils have compact, sandy or silty textures that hinder drainage and root growth.
  - These soils are nutrient-poor, with pale or cracked surfaces, making them infertile for most crops.
  - It is known by various local names such as reh, kallar, usar, thur, rakar, karl, and chopan.
- **Crops:** Barley, sugar beet, cotton, wheat, rice, sorghum, maize, sunflower, potato, citrus fruits, pomegranate, peas, lentils, cowpea etc.

‹ ›

Soils of India Significance

Enquire Now

India's diverse topography, from the Himalayan mountains to southern plateaus, fosters a wide range of soils. These varied soils are significant in sustaining diverse agricultural activities, maintaining ecological balance and supporting the economy.

- **Agricultural Productivity**: Diverse soil types support a wide range of agricultural activities, directly sustaining over 40% of India's population. Alluvial soils, in particular, offer highly productive lands essential for farming.
- **Ecosystem Health**: Soils are vital for maintaining ecological balance, enabling plant growth that supports animal life and biodiversity. Each soil type fosters specific ecosystems critical for environmental stability.
- **Economic Foundation**: Agriculture, dependent on soil fertility, is the backbone of India's economy, supporting 65-70% of the population and influencing food production and economic stability.
- **Carbon Sequestration:** Soils are crucial for carbon sequestration, aiding in climate change mitigation. Properly managed soils can capture large amounts of carbon dioxide from the atmosphere, helping to regulate the climate.
- **Resilience Against Natural Disasters:** Healthy soils strengthen landscape resilience to natural disasters like floods and droughts. They help ecosystems recover by retaining moisture and minimizing erosion during heavy rainfall.

Don't miss Vajiram & Ravi updates:
Add Vajiram & Ravi as a preferred source on Google
Add →

Soils of India UPSC PYQs

Question 1: The black cotton soil of India has been formed due to the weathering of **(UPSC Prelims 2021)**

(a) Brown forest soil

(b) Fissure volcanic rock

(c) Granite and schist

(d) Shale and limestone

Answer: (b)

Question 2: With reference to agricultural soils, consider the following statements : **(UPSC Prelims 2018)**

1. High content of organic matter in soil drastically reduces its water holding capacity.
2. Soil does not play any role in the sulphur cycle.

Enquire Now

3. Irrigation over a period of time can contribute to the salinization of some agricultural lands.

Which of the statements given above is/are correct?

(a) 1 and 2 only

(b) 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Answer: (b)

Question 3: Which of the following statements regarding laterite soils of India are correct? **(UPSC Prelims 2013)**

1. They are generally red in colour.
2. They are rich in nitrogen and potash.
3. They are well-developed in Rajasthan and UP.
4. Tapioca and cashew nuts grow well on these soils.

Select the correct answer using the codes given below.

(a) 1, 2 and 3

(b) 2, 3 and 4

(c) 1 and 4

(d) 2 and 3 only

Answer: (c)

Other Related Posts	
Chota Nagpur Plateau	Interior of the Earth
Malwa Plateau	Continental Drift Theory
Soils of India	Paleomagnetism
Largest State in India	Convectional Current Theory
Godavari River System	Ts

Enquire Now

Big Bang Theory	Western Disturbances in India
Latitude and Longitude	The Universe
Asteroid	

Latest UPSC Exam 2026 Updates

Last updated on April, 2026

→ UPSC Final Result 2025 is now out.

→ UPSC has released UPSC Toppers List 2025 with the Civil Services final result on its official website.

→ Anuj Agnihotri secured AIR 1 in the UPSC Civil Services Examination 2025.

→ UPSC Marksheet 2025 is now out.

→ UPSC Notification 2026 & UPSC IFoS Notification 2026 is now out on the official website at upsconline.nic.in.

→ **UPSC Calendar 2026** has been released.

→ Check out the latest UPSC Syllabus 2026 here.

→ UPSC Prelims 2026 will be conducted on **24th May, 2026** & **UPSC Mains 2026** will be conducted on **21st August 2026.**

→ The UPSC Selection Process is of 3 stages-Prelims, Mains and Interview.

→ Prepare effectively with Vajiram & Ravi's UPSC Prelims Test Series 2026 featuring full-length mock tests, detailed solutions, and performance analysis.

→ Enroll in Vajiram & Ravi's UPSC Mains Test Series 2026 for structured answer writing practice, expert evaluation, and exam-oriented feedback.

→ Join Vajiram & Ravi's Best UPSC Mentorship Program for personalized guidance, strategy planning, and one-to-one support from experienced mentors.

→ Shakti Dubey secures AIR 1 in UPSC CSE Exam 2024.

Enquire Now

→ Also check Best UPSC Coaching in India

Soils of India FAQs

Q1. What are the 7 types of soil in India? +

Q2. What are the division of soil? +

Q3. Who is the father of soil taxonomy? +

Q4. What is called loam? +

Q5. Which soil is known as regur? +

Tags: quest soils of india

Previous Article
NHRC - National Human Rights Commission, Powers, Functions

Next Article
President of India

Vajiram Editor

At Vajiram & Ravi, our team includes subject experts who have appeared for the UPSC Mains and the Interview stage. With their deep understanding of the exam, they create content that is clear, to the point, reliable, and helpful for aspirants.Their aim is to make even difficult topics easy to understand and directly useful for your UPSC preparation—whether it's for Current Affairs, General Studies, or Optional subjects. Every note, article, or test is designed to save your time and boost your performance.

Related Posts

Enquire Now

Swarajists and No-Changers

Vajiram Editor | April 11, 2026

Round Table Conferences (RTC)

Vajiram Editor | April 11, 2026

Chauri Chaura Incident

Vajiram Editor | April 11, 2026

Our Latest UPSC Courses

SC Mentorship Program
5000
nroll Now

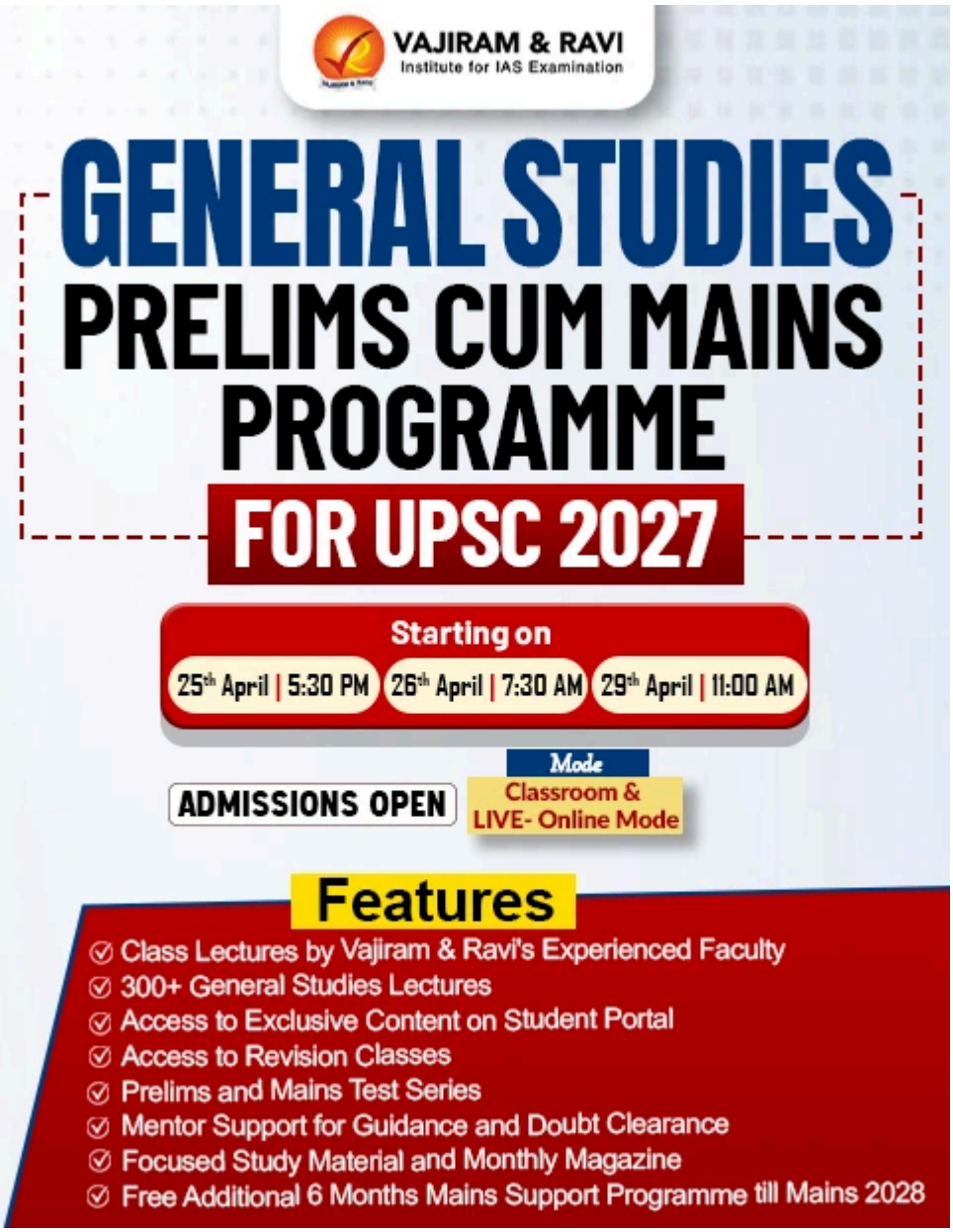
UPSC Sureshot Mains Test Series
₹19000
Enroll Now

Prelims Powerup Test Ser
₹8500
Enroll Now

Enquire Now

VAJIRAM AND RAVI'S
UPSC CSE 2025 TOPPERS
Our Top Achievers
Top 10 Out Of 10 Rankers From
Vajiram & Ravi's Various Courses
RANK 1
Anuj Agnihotri
RANK 2
Rajeshwari Suve M
PowerUp Prelims Test Series, Sure Shot Mains Test Series, IGP
RANK 3
Akansh Dhull
General Studies Classroom Programme
RANK 4
Raghav Jhunjhunwala
IGP
RANK 5
Ishan Bhatnagar
Sociology Optional Crash Course, Sociology Enrichment Course, Prelims CAMP, IGP
RANK 6
Zinnia Aurora
GS Ethics Module, PowerUp Prelims Test Series, IGP
RANK 7
A R Rajah Mohaideen
IGP
RANK 8
Pakshal Secretry
Mains CAMP, IGP
RANK 9
Astha Jain
IGP
RANK 10
Ujjwal Priyank
IGP

Enquire Now



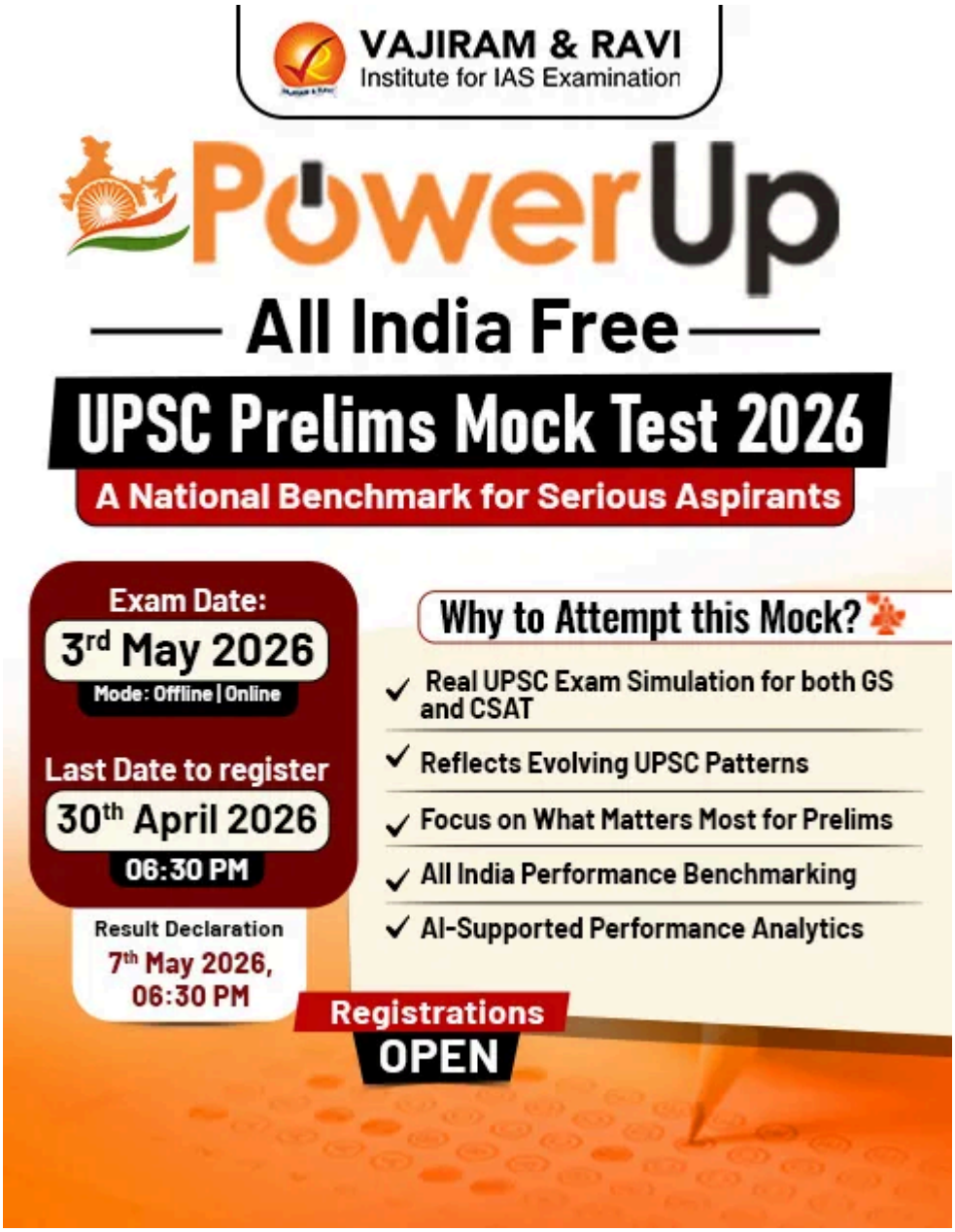
Recent Posts

- UPSC Prelims 2026, Exam Date, Syllabus, Eligibilty, Admit Card
- UPSC Calendar 2026, Check CSE Exam Date, Download PDF
- UPSC Cut Off 2025 Out, Check Prelims, Mains, & Final Cut off Now
- UPSC Application Form 2026 Out, Last Date, Steps to Apply, Fees
- UPSC Age Limit 2026, Eligibility, Qualification, Attempts

Enquire Now

VAJIRAM & RAVI
Institute for IAS Examination
STEP UP
MENTORSHIP PROGRAMME 2027
Available in
English Medium
&
Hindi Medium
Starting on
1st May 2026
Mode
Offline & Online
ADMISSIONS OPEN
For aspirants tired of studying hard but not improving
Individuals fail. Systems don't.
Prepare inside a structure, not alone.
Learn how toppers think — not just what they read.

Enquire Now



UPSC Exam 2026

- UPSC Notification 2026
- UPSC Calendar 2026
- UPSC Prelims 2026
- UPSC Mains 2026
- UPSC Apply Online
- UPSC Age Limit
- UPSC Syllabus
- UPSC Mains Syllabus
- UPSC Books
- NCERT Books for UPSC
- UPSC Optional Subjects List

Enquire Now

IAS Salary

UPSC Previous Year Question Papers

UPSC Admit Card

UPSC Prelims Result

UPSC Mains Result

UPSC Final Result

UPSC Cut Off

IAS vs IPS

UPSC Courses

UPSC Foundation Course

UPSC Prelims Test Series 2026

UPSC Mains Test Series 2026

CSAT Course

UPSC Mentorship Program 2027

UPSC Optional Subject Coaching

UPSC Optional Test Series 2026

UPSC Interview Guidance Program

About Vajiram & Ravi

About Us

Disclaimer

Contact Us

Terms and Conditions

Privacy Policy

Careers

FAQs

About UPSC Exam

UPSC Notification 2026

UPSC Syllabus

UPSC Syllabus in Hindi

UPSC Exam Pattern

IAS Salary

UPSC Age Limit

Enquire Now

UPSC Calendar 2026
UPSC Interview
UPSC Optional Subjects
UPSC Selection Process
UPSC CSAT Syllabus
UPSC Books
UPSC Previous Year Papers
UPSC Final Result 2025
UPSC Cut Off
UPSC Current Affairs
UPSC Answer Key
UPSC Admit Card

UPSC Courses

UPSC GS Online Coaching 2026
GS 2 Years Foundation Course
UPSC CSAT Coaching
UPSC Prelims Test Series
UPSC Mains Test Series
UPSC Mentorship Program
UPSC Optional Coaching
UPSC Optional Test Series
UPSC IFoS Coaching
UPSC Interview Guidance IGP
UPSC Online Coaching

Install App

Connect on Social Media

© 2026 Vajiram & Ravi • All rights reserved.

Enquire Now